

Motivation

Every call holds a story—missed or maximized. **Agentic AI for Conversation Analysis** turns passive recordings into strategic assets by using AI agents to decode, summarize, and surface actionable insights instantly. This results in faster decisions, smarter sales, and elevated customer experience.

Abstract

- Agentic AI system engineered to elevate sales performance through intelligent analysis of call center conversations.
- By automating the retrieval and transcription of call recordings, the system applies large language models to extract context-aware insights, enabling task execution and structured output generation.
- The results are visualized through dashboard, empowering stakeholders with real-time analytics on sales effectiveness and team impact.
- This seamless integration of speech processing, reasoning, and data visualization offers a powerful, scalable tool for data-driven decision-making in modern sales operations.

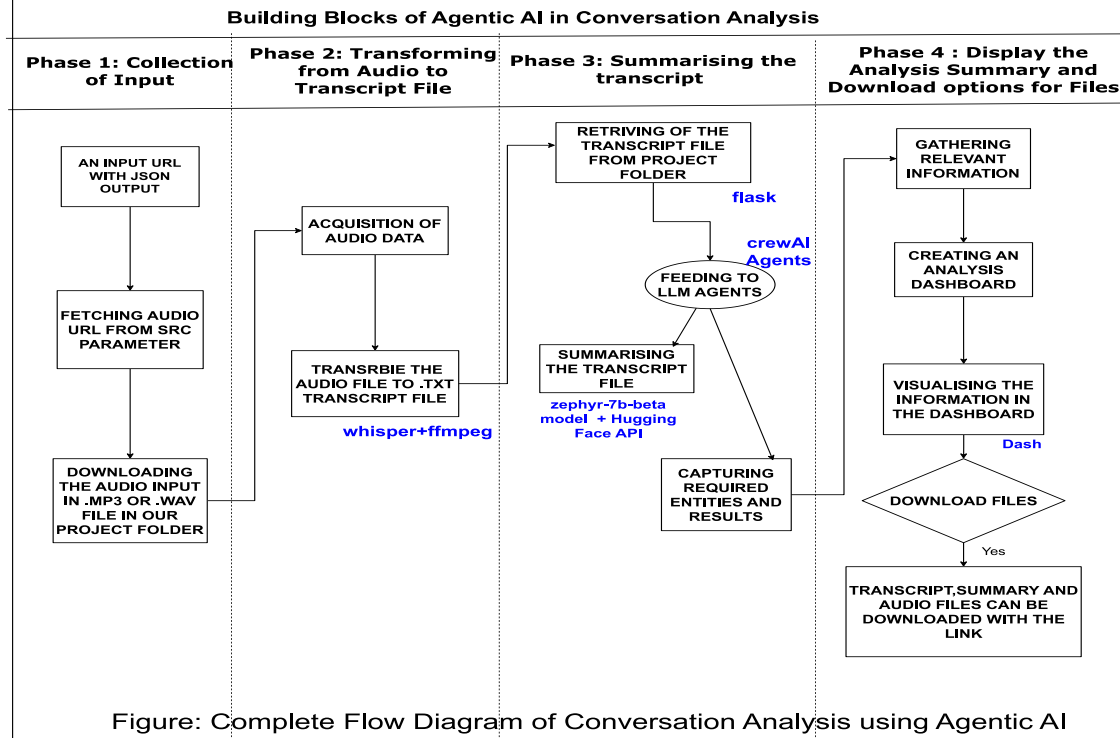
Tools



Conclusion

- Implemented a CrewAI-driven workflow that automates audio retrieval, Whisper transcription, and HuggingFace analysis—delivering seamless, dashboard-ready insights from raw speech.
- This agentic pipeline transforms call center conversations into strategic sales intelligence with precision, scalability, and real-time impact.

Architecture



Results

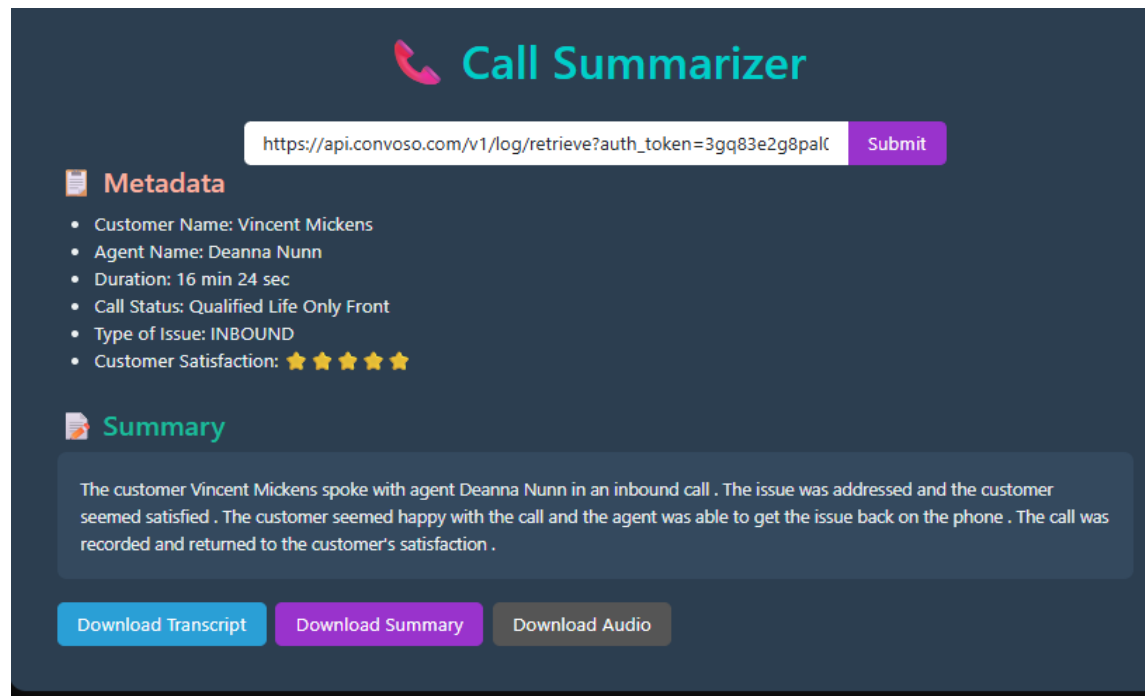


Figure: Dashboard for Analysis of Audio from Input URL.

Future work

Agentic AI doesn't stop here; we can extend our Agent capability to next level. Planning for the following integrations.

- Real-Time Agent Interaction
- Integrating with Model Context protocol (MCP).
- Multimodal Insight Integration
- Adaptive Feedback & Coaching

References

- Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). Robust Speech Recognition via Large-Scale Weak Supervision. arXiv. Retrieved from <https://arxiv.org/abs/2212.04356>
- CrewAI. Crafting effective agents. <https://docs.crewai.com/en/guides/agents/crafting-effective-agents>, 2024. Accessed: 2025-06-26.